

11.2 Spectral Decomposition



Core skills you will master in this section:

- ▶ A real symmetric matrix: satisfies $\mathbf{A} = \mathbf{A}^T$, with all elements being real numbers.
- ▶ The spectral decomposition of a real symmetric matrix can be expressed as "rotate $\rightarrow$ scale $\rightarrow$ rotate."
- ▶ Spectral decomposition finds a new coordinate system that sets the ellipse upright.
- ▶ Unit circle $\rightarrow$ rotated ellipse, reflecting the geometric meaning of eigenvalues.
- ▶ A square matrix is invertible only when all of its eigenvalues are nonzero.

The previous section mentioned that the eigenvalue decomposition of a real symmetric matrix is called spectral decomposition. Simply put, a real symmetric matrix is a symmetric matrix all of whose elements are real numbers. As mentioned earlier in this book, if matrix $\mathbf{A}$ is a symmetric matrix, it satisfies $\mathbf{A} = \mathbf{A}^T$. Geometrically, a symmetric matrix is mirror-symmetric about its main diagonal.

⚠ Note that unless stated otherwise, this book assumes by default that all matrix elements are real numbers.

Let's first review (learn) the concepts of real numbers, imaginary numbers, and complex numbers.

Real Numbers, Imaginary Numbers, Complex Numbers

You should already be familiar with the concept of real numbers. Real numbers (real number) include rational numbers (which can be expressed as the ratio of two integers, such as the repeating decimal $1/3$) and irrational numbers (which cannot be expressed as the ratio of two integers, such as the non-repeating decimal π).

As shown in Figure 1, every real number corresponds to a point on the number line; conversely, every point on the number line also corresponds to a real number. In other words, the rational number $1/3$ and the irrational number π each have their own position on the number line.

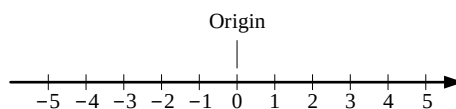


Figure 1. The Real Number Line

The opposite of a real number is an imaginary number (imaginary number). The definition of an imaginary number comes from extending the square-root operation to negative numbers. i is generally defined as the imaginary unit (imaginary unit), $i^2 = -1$.

Putting real numbers and imaginary numbers together, we get complex numbers (complex number), such as $a + bi$ (where a and b are both real numbers).

⚠ Note that the reason we bring up the concepts of real numbers, imaginary numbers, and complex numbers is that, as you will soon see, complex numbers can appear among the eigenvalues and eigenvectors of an eigenvalue decomposition. However, operations involving complex matrices are not a topic covered in this book.

Spectral Decomposition of a Real Symmetric Matrix

Given an $n \times n$ real symmetric matrix A , its eigenvalue decomposition can be written as

$$A = V\Lambda V^T$$

where V is an orthogonal matrix, satisfying $VV^T = V^T V = I$.

V 's column vectors are the eigenvectors of matrix A . Λ is a diagonal matrix, and Λ 's main diagonal elements are A 's eigenvalues.

We call the eigenvalue decomposition of a real symmetric matrix the spectral decomposition (spectral decomposition). You will soon see where the word "spectral" comes from.

⚠ Note that whenever this book refers to a symmetric matrix, it is assumed by default to be a real symmetric matrix.

The Example from the Previous Section

Continuing the example from the previous section, the eigenvalue decomposition (spectral decomposition) of the 2×2 real symmetric matrix A is:

$$A = V\Lambda V^T = \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_V \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}}_\Lambda \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{V^T} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

Applying A to the two-dimensional column vector x corresponds to the matrix multiplication

$$Ax = y$$

A can be viewed as a composite geometric transformation. Based on the equation above, we can break it down into three sequential actions

$$Ax = V\Lambda V^T x = y$$

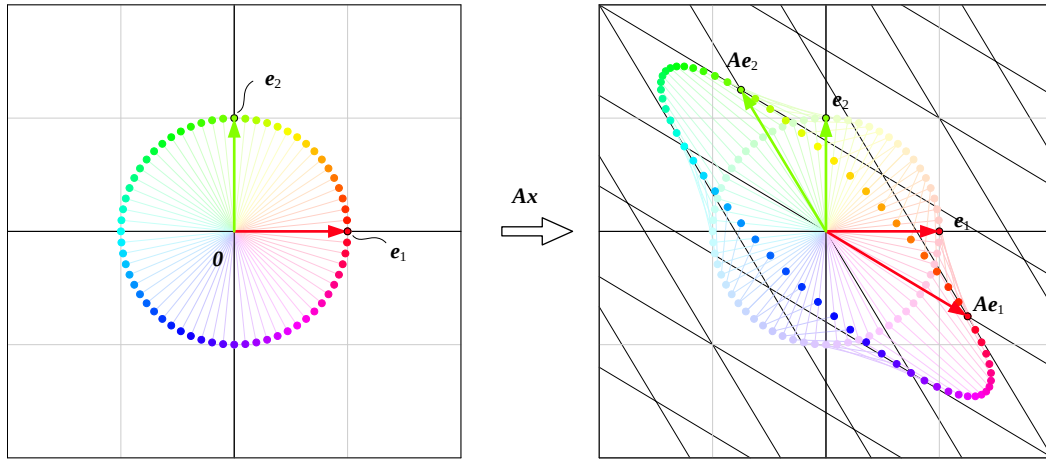
Figure 2 shows the geometric transformation that A applies to the two unit vectors e_1 and e_2 .

A 's geometric operation on e_1 gives

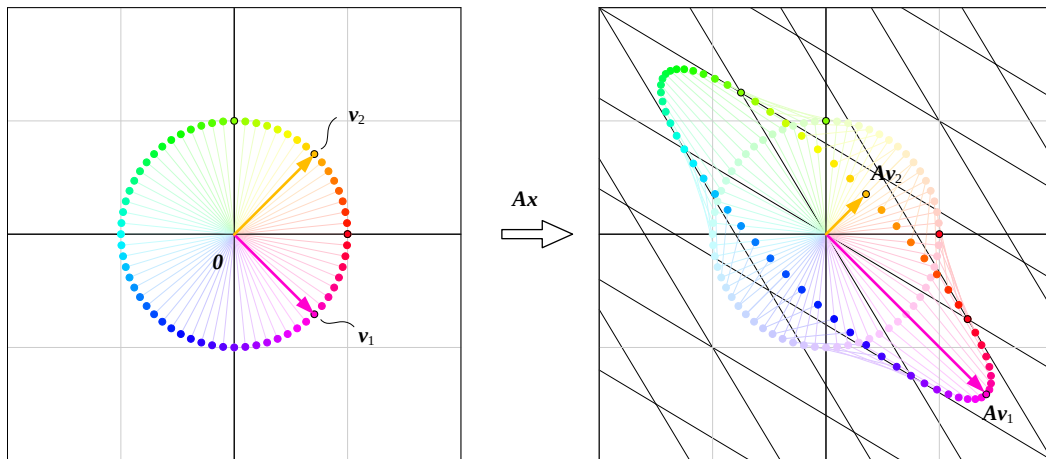
$$Ae_1 = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5/4 \\ -3/4 \end{bmatrix}$$

A 's geometric operation on e_2 gives

$$Ae_2 = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 5/4 \end{bmatrix}$$

Figure 2. Linear Transformation of Matrix A on e_1 and e_2

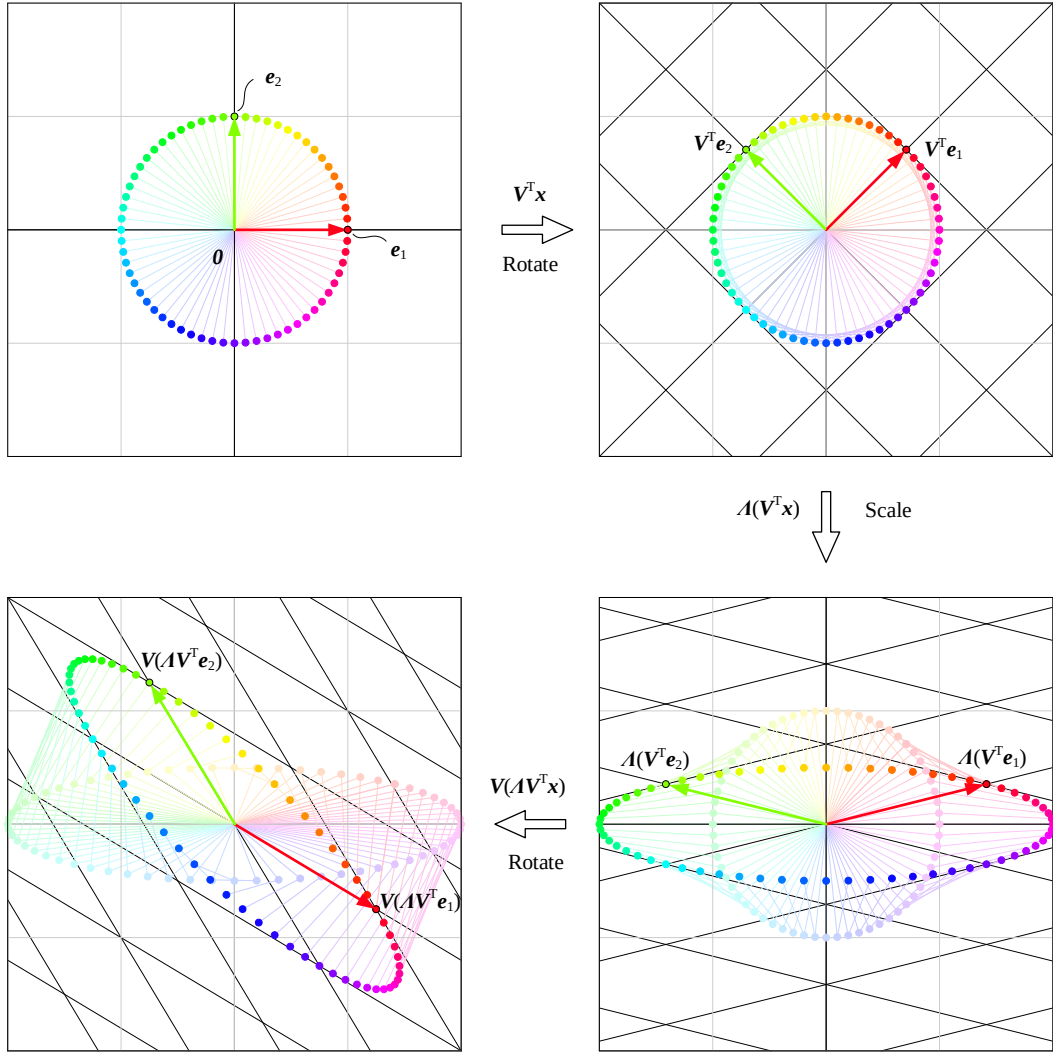
? Please write out the matrix multiplication corresponding to Figure 3.

Figure 3. Linear Transformation of Matrix A on v_1 and v_2

Next, let's look at the step-by-step effect of WV^T 's action.

e_1 and e_2 : Rotate $\rightarrow$ Scale $\rightarrow$ Rotate

WV^T 's step-by-step geometric operations on e_1 and e_2 are shown specifically in Figure 4. Let's analyze them step by step below.

Figure 4. Linear Transformation of Matrix VAV on e_1 and e_2

First, rotate using the rotation matrix V

$$V^T = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

⚠ As mentioned earlier in this book, the identity matrix, rotation matrices, permutation matrices, reflection matrices, and any combination of them in any order, are all orthogonal matrices. For convenience of discussion, in this chapter we simplify the geometric operation of an orthogonal matrix to a rotation.

For example, the result of applying V to e_1 is

$$V^T e_1 = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

V 's rotation result on e_2 is

$$V^T e_2 = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

It is easy to compute that the determinant of V is 1; clearly, rotation does not change area.

In the equation above, the scaling matrix A is

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

The geometric effect of this diagonal matrix is to stretch the horizontal axis by a factor of 2 and shrink the vertical axis to 1/2.

Coincidentally, the determinant of this diagonal matrix is also 1. This means that in Figure 4, the area of the ellipse is the same as that of the unit circle. The determinant of matrix A equals the product of all its eigenvalues.

A 's scaling result on $V^T e_1$ is

$$A(V^T e_1) = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ \sqrt{2}/4 \end{bmatrix}$$

A 's scaling result on $V^T e_2$ is

$$A(V^T e_2) = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} -\sqrt{2} \\ \sqrt{2}/4 \end{bmatrix}$$

Finally, rotate using the following V

$$V = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

V 's rotation result on $AV^T e_1$ is

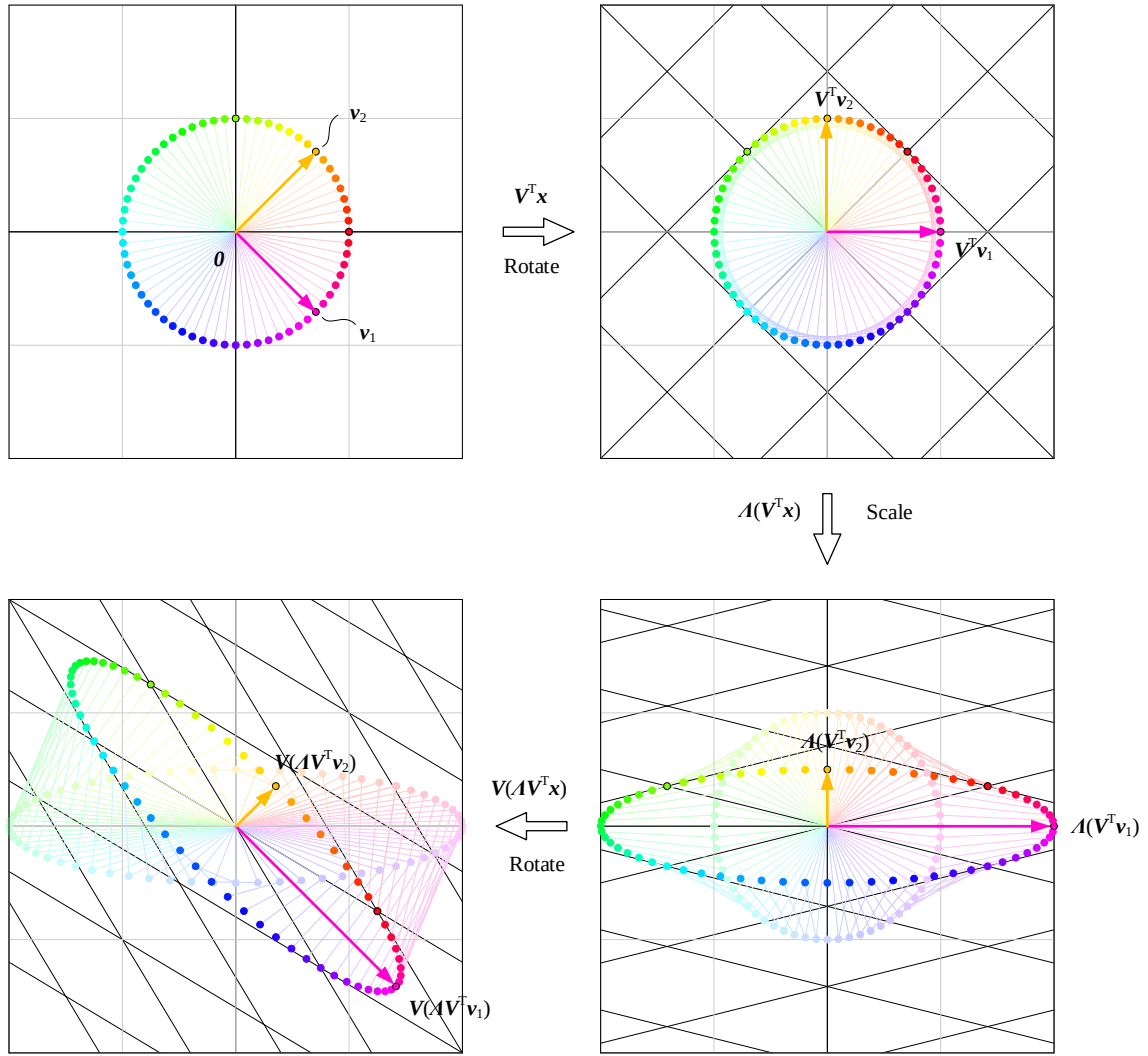
$$V(AV^T e_1) = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} \sqrt{2} \\ \sqrt{2}/4 \end{bmatrix} = \begin{bmatrix} 5/4 \\ -3/4 \end{bmatrix}$$

V 's rotation result on $AV^T e_2$ is

$$V(AV^T e_2) = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} -\sqrt{2} \\ \sqrt{2}/4 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 5/4 \end{bmatrix}$$

v_1 and v_2 : Rotate $\rightarrow$ Scale $\rightarrow$ Rotate

Next, let's look again at VAV^T 's action on the eigenvectors v_1 and v_2 .

Figure 5. Linear Transformation of Matrix $A(V^T)$ on v_1 and v_2

For example, the result of applying V^T to v_1 is

$$V^T v_1 = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

V^T 's rotation result on v_2 is

$$V^T v_2 = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

we actually get the basis vectors of the standard orthonormal basis, e_1 and e_2 ? This is clearly not a coincidence!

Expanding the orthogonal matrix V into column-vector form, its two column vectors can be written as

$$\begin{aligned} V^T v_1 &= [v_1 \ v_2]^T v_1 = \begin{bmatrix} v_1^T \\ v_2^T \end{bmatrix} v_1 = \begin{bmatrix} v_1^T v_1 \\ v_2^T v_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ V^T v_2 &= [v_1 \ v_2]^T v_2 = \begin{bmatrix} v_1^T \\ v_2^T \end{bmatrix} v_2 = \begin{bmatrix} v_1^T v_2 \\ v_2^T v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}$$

Next, let's look at A 's scaling effect.

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Λ 's scaling result on $V^T v_1$ is

$$\Lambda(V^T v_1) = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

Λ 's scaling result on $V^T v_2$ is

$$\Lambda(V^T v_2) = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$$

As shown in Figure 5, the unit circle becomes an upright ellipse.

Finally, rotate using V . V 's rotation result on $\Lambda V^T v_1$ is

$$V(\Lambda V^T v_1) = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ -\sqrt{2} \end{bmatrix} = 2 \times \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix} = 2v_1$$

V 's rotation result on $\Lambda V^T v_2$ is

$$V(\Lambda V^T v_2) = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 0 \\ 1/2 \end{bmatrix} = \begin{bmatrix} \sqrt{2}/4 \\ \sqrt{2}/4 \end{bmatrix} = 0.5 \times \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} = 0.5v_2$$

After the upright ellipse is rotated, we get the rotated ellipse.

Viewing the Ellipse in $[v_1, v_2]$

The column vectors of the orthogonal matrix V form a standard orthonormal basis. Figure 6 shows the ellipse viewed in $[v_1, v_2]$. It is easy to see that, geometrically, spectral decomposition is equivalent to helping us find a coordinate system that sets the ellipse upright!

λ_1 and λ_2 are the semi-axis lengths of the ellipse.



This point will be used in Principal Component Analysis (PCA), which will be introduced in the next chapter.

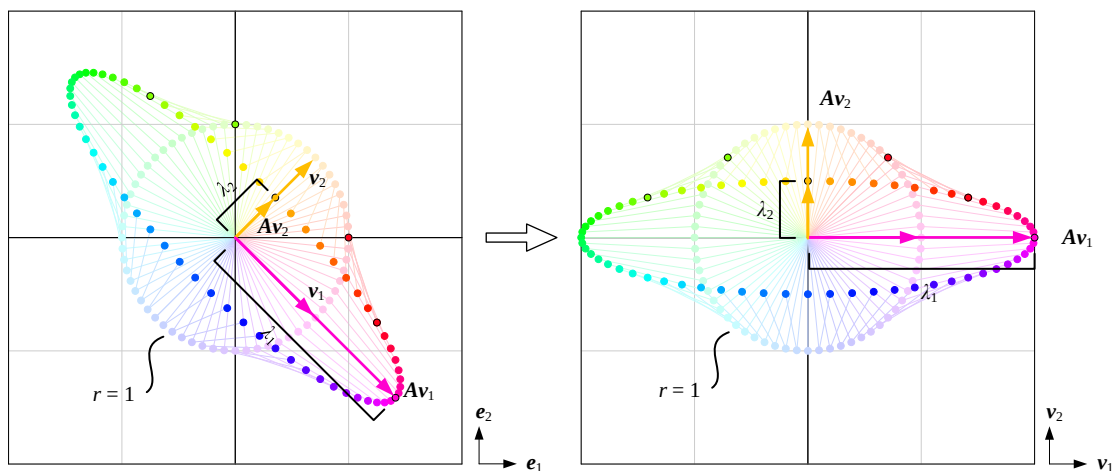


Figure 6. Viewing the Ellipse in $[e_1, e_2]$ and $[v_1, v_2]$, Respectively

This ellipse must have an analytic equation — so what is the relationship between this equation and matrix A ? How do we find the lengths of the ellipse's semi-major and semi-minor axes? Let's answer these questions below.

The Ellipse

The ellipse (ellipse) is a very important curve in geometry, and is a type of conic section (conic section).

Simply put, an ellipse is the locus of points in a plane whose distances to two fixed points sum to a constant value. These two fixed points are called the foci (foci), usually denoted F_1 and F_2 .

As shown in Figure 7, any point P on the ellipse satisfies

$$PF_1 + PF_2 = 2a$$

where $a > c > 0$; c is the distance from a focus to the center of the ellipse.

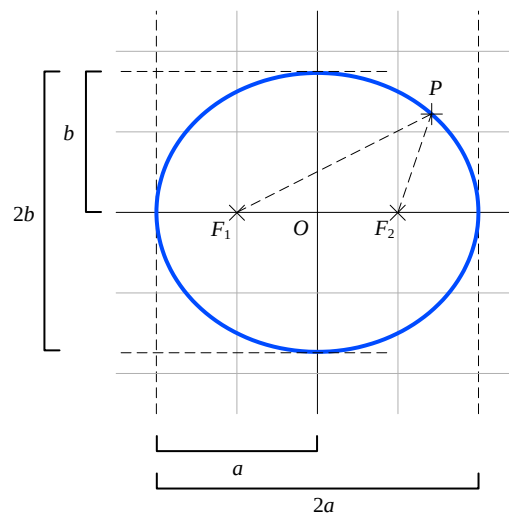


Figure 7. An Ellipse Centered at the Origin with Foci on the Horizontal Axis

Figure 7 — in this figure, the equation of the ellipse centered at the origin with foci on the x_1 -axis (horizontal axis) is

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$$

where $a > b > 0$.

As shown in Figure 7, the major axis (major axis) passes through both foci and has length $2a$. The length a is the semi-major axis (semi-major axis);

the minor axis (minor axis) is perpendicular to the major axis and has length $2b$. The length b is the semi-minor axis (semi-minor axis).

We call an ellipse whose major and minor axes are parallel to the horizontal and vertical axes an upright ellipse; when the major and minor axes are rotated about the center of the ellipse, the result is a rotated ellipse.

Looking back at the right panel of Figure 6, we find that once the ellipse is set upright, the semi-major axis length is λ_1 and the semi-minor axis length is λ_2 ; the analytic equation of the ellipse is

$$\frac{x_1^2}{\lambda_1^2} + \frac{x_2^2}{\lambda_2^2} = 1$$

Written in matrix multiplication form

We can easily write the equation above in matrix multiplication form

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1/a^2 & 0 \\ 0 & 1/b^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 1$$

Based on this, the equation above can also be written in matrix multiplication form, namely

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1/\lambda_1^2 & 0 \\ 0 & 1/\lambda_2^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 1$$

How do we get from the unit circle to this ellipse, step by step?

Unit Circle $\rightarrow$ Ellipse

Assume the point on the unit circle corresponds to the unit vector

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

This unit vector's length (modulus, magnitude, L^2 norm, Euclidean norm) is 1, corresponding to the equation

$$\|\mathbf{z}\|^2 = \mathbf{z}^T \mathbf{z} = 1$$

Expanding, we get

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}^T \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = z_1^2 + z_2^2 = 1$$

After matrix $\mathbf{A}$ acts on $\mathbf{z}$, we get $\mathbf{x}$, corresponding to the matrix multiplication

$$\mathbf{A}\mathbf{z} = \mathbf{x}$$

Since matrix $\mathbf{A}$ is invertible, therefore

$$\mathbf{z} = \mathbf{A}^{-1}\mathbf{x}$$

Substituting this into the equation above, we get

$$(\mathbf{A}^{-1}\mathbf{x})^T \mathbf{A}^{-1}\mathbf{x} = 1$$

Simplifying

$$\mathbf{x}^T (\mathbf{A}^{-1})^T \mathbf{A}^{-1} \mathbf{x} = 1$$

The inverse of matrix $\mathbf{A}$ is

$$\mathbf{A}^{-1} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}^{-1} = \begin{bmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{bmatrix}$$

Substituting into the equation above and simplifying, we get the analytic equation of the rotated ellipse

$$2.125x_1^2 + 3.75x_1x_2 + 2.125x_2^2 = 1$$

It is not so easy to directly obtain the lengths of the ellipse's semi-major and semi-minor axes from this equation.

Let's take a different approach. Since $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, $\mathbf{A}$'s inverse can be written as

$$\mathbf{A}^{-1} = (\mathbf{V}^T)^{-1} \mathbf{\Lambda}^{-1} \mathbf{V}^{-1} = \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T$$

Substituting this into the equation above, we get

$$\mathbf{x}^T (\mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T)^T \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T \mathbf{x} = 1$$

Simplifying

$$\mathbf{x}^T \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T \mathbf{x} = 1$$

we get

$$(\mathbf{V}^T \mathbf{x})^T \mathbf{\Lambda}^{-2} (\mathbf{V}^T \mathbf{x}) = 1$$

Let $\mathbf{y} = \mathbf{V}^T \mathbf{x}$, the equation above can be written as

$$\mathbf{y}^T \mathbf{\Lambda}^{-2} \mathbf{y} = 1$$

that is

$$\begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 1/\lambda_1^2 & 0 \\ 0 & 1/\lambda_2^2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 1$$

Simplifying, we get

$$\frac{y_1^2}{\lambda_1^2} + \frac{y_2^2}{\lambda_2^2} = 1$$

This is the upright-ellipse form we wanted.



In addition to the ellipse, conic sections also include the hyperbola, parabola, and so on. What relationship do they have with linear algebra? This is a topic that Chapter 13 of this book will discuss.

The More General Spectral Decomposition

Now that we understand the spectral decomposition of the 2×2 real symmetric matrix $\mathbf{A}$, let's talk about the more general spectral decomposition.

As shown in Figure 8, the $n \times n$ real symmetric matrix $\mathbf{A}$, after spectral decomposition, gives $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$. $\mathbf{V}$ is an orthogonal matrix, satisfying $\mathbf{V}\mathbf{V}^T = \mathbf{V}^T\mathbf{V} = \mathbf{I}$.

$\mathbf{\Lambda}$ is a diagonal matrix, with eigenvalues on its main diagonal.

Figure 8, the main diagonal elements of $\mathbf{A}$ are arranged in decreasing order from top-left to bottom-right.

For convenience of discussion, geometrically, $\mathbf{V}$ completes the rotation in $\mathbb{R}^n$, and $\mathbf{A}$ completes the scaling in $\mathbb{R}^n$.

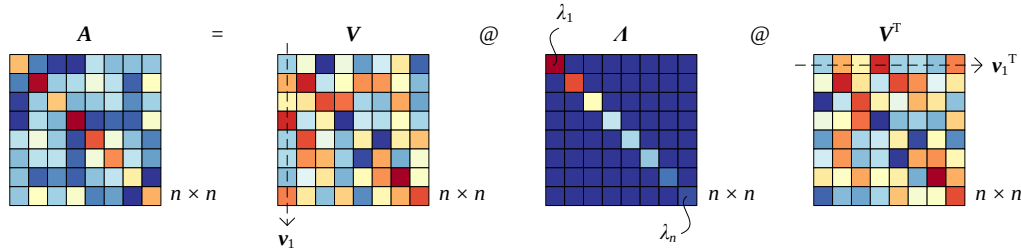


Figure 8. Spectral Decomposition of the $n \times n$ Real Symmetric Matrix $\mathbf{A}$

The Second Perspective on Matrix Multiplication

Using the second perspective on matrix multiplication — the outer-product perspective — we can write $\mathbf{A} = \mathbf{V}\mathbf{A}\mathbf{V}^T$ as

$$\mathbf{A} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix}$$

$$= \lambda_1 \mathbf{v}_1 \mathbf{v}_1^T + \lambda_2 \mathbf{v}_2 \mathbf{v}_2^T + \cdots + \lambda_n \mathbf{v}_n \mathbf{v}_n^T$$

This way, we write matrix $\mathbf{A}$ as a sum of n matrices, as shown in Figure 9. The larger the eigenvalue, the more "vivid" the color of the corresponding sub-figure. As long as λ_i is nonzero, $\lambda_i \mathbf{v}_i \mathbf{v}_i^T$ is a rank-one matrix; otherwise, if λ_i is 0, $\lambda_i \mathbf{v}_i \mathbf{v}_i^T$ is the zero matrix.

Performing spectral decomposition on a real symmetric matrix is like "breaking it apart" into the sub-figure matrices in Figure 9. These matrices are like the orderly sequence of color bands revealed when white light is split by a prism — these different colors of light represent different wavelength components.

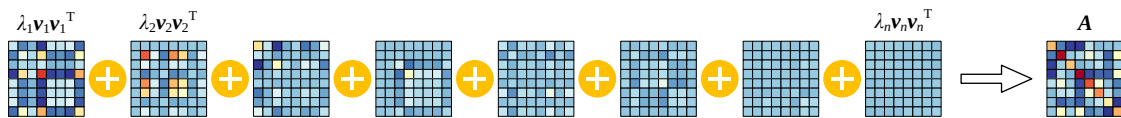
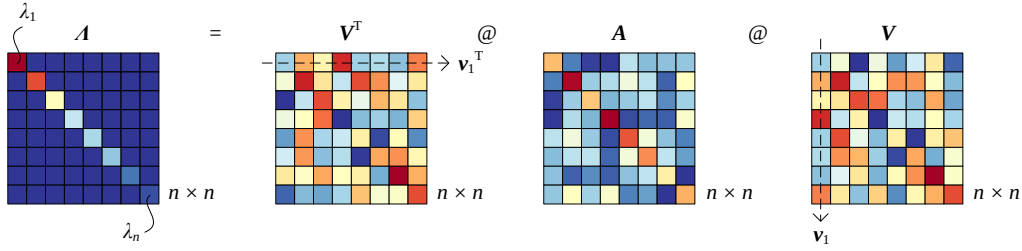


Figure 9. Sum of n Matrices Gives $\mathbf{A}$

Diagonalization

Clearly the diagonal matrix $\mathbf{A}$ is invertible. As shown in Figure 10, the diagonalization of the real symmetric matrix $\mathbf{A}$ corresponds to the matrix multiplication

$$\mathbf{V}^T \mathbf{A} \mathbf{V} = \mathbf{A}$$

Figure 10. Diagonalization of the $n \times n$ Real Symmetric Matrix $\mathbf{A}$

Using the first perspective on matrix multiplication to expand the equation above, we get

$$\begin{aligned}
 \mathbf{V}^T \mathbf{A} \mathbf{V} &= \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix} \mathbf{A} [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n] \\
 &= \begin{bmatrix} \mathbf{v}_1^T \mathbf{A} \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{A} \mathbf{v}_2 & \cdots & \mathbf{v}_1^T \mathbf{A} \mathbf{v}_n \\ \mathbf{v}_2^T \mathbf{A} \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{A} \mathbf{v}_2 & \cdots & \mathbf{v}_2^t \mathbf{A} \mathbf{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_n^T \mathbf{A} \mathbf{v}_1 & \mathbf{v}_n^T \mathbf{A} \mathbf{v}_2 & \cdots & \mathbf{v}_n^T \mathbf{A} \mathbf{v}_n \end{bmatrix} \\
 &= \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}
 \end{aligned}$$

Let's first look at the main-diagonal matrix multiplication in the equation above.

Take $\mathbf{v}_1^T \mathbf{A} \mathbf{v}_1 = \lambda_1$ as an example. Since $\mathbf{A} \mathbf{v}_1 = \lambda_1 \mathbf{v}_1$, and $\mathbf{v}_1$ is a unit vector, $\mathbf{v}_1^T \mathbf{A} \mathbf{v}_1 = \lambda_1$ can be written as

$$\mathbf{v}_1^T \mathbf{A} \mathbf{v}_1 = \lambda_1 \mathbf{v}_1^T \mathbf{v}_1 = \lambda_1 \|\mathbf{v}_1\|_2^2 = \lambda_1$$

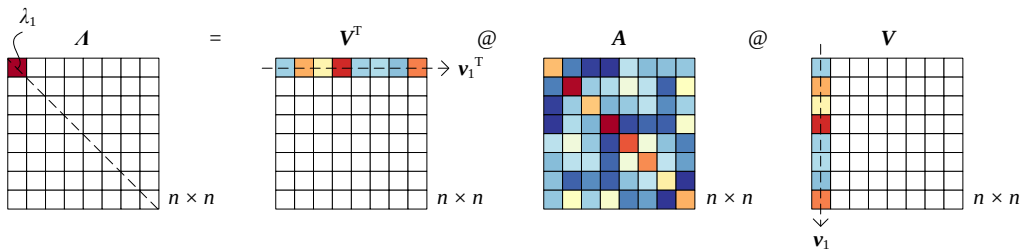
The equation above is a special form of the Rayleigh quotient (Rayleigh quotient), used to measure the scaling effect on vector $\mathbf{v}_1$ under the action of $\mathbf{A}$.

Given a real symmetric matrix $\mathbf{A}$, $\mathbf{A}$'s Rayleigh quotient is defined as:

$$R(\mathbf{x}) = \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$$



Chapter 13, Section 3 of this book will introduce the Rayleigh quotient in detail.

Figure 11. Diagonalization of the $n \times n$ Real Symmetric Matrix $\mathbf{A}$, Main Diagonal Elements

Now let's look at the off-diagonal multiplication in the equation above. Take $\mathbf{v}_2^T \mathbf{A} \mathbf{v}_1 = 0$ as an example. Since $\mathbf{V}$'s column vectors are pairwise orthogonal, this expression can be written as

$$\mathbf{v}_2^T \mathbf{A} \mathbf{v}_1 = \lambda_1 \mathbf{v}_2^T \mathbf{v}_1 = 0$$

The equation above tells us that under the specific linear transformation of $\mathbf{A}$, the eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ still remain orthogonal.

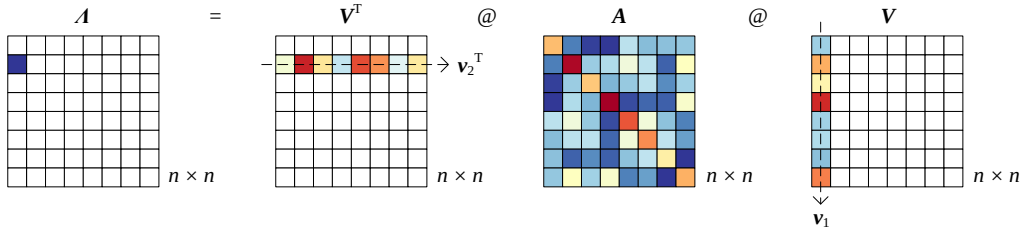


Figure 12. Diagonalization of the $n \times n$ Real Symmetric Matrix $\mathbf{A}$, Off-Diagonal Elements

In addition, since $\mathbf{A}$ is a real symmetric matrix, the equation above can also be transposed to give

$$\mathbf{v}_1^T \mathbf{A} \mathbf{v}_2 = \lambda_1 \mathbf{v}_1^T \mathbf{v}_2 = 0$$

The Inverse Matrix

Only when all eigenvalues are nonzero does the real symmetric matrix $\mathbf{A}$ have an inverse, namely

$$\mathbf{A}^{-1} = (\mathbf{V} \mathbf{A} \mathbf{V}^T)^{-1} = (\mathbf{V}^T)^{-1} \mathbf{A}^{-1} \mathbf{V}^{-1} = \mathbf{V} \mathbf{A}^{-1} \mathbf{V}^T = \mathbf{V} \begin{bmatrix} 1/\lambda_1 & & \\ & 1/\lambda_2 & \\ & & \ddots \\ & & & 1/\lambda_n \end{bmatrix} \mathbf{V}^T$$

The equation above is equivalent to the spectral decomposition of the inverse matrix $\mathbf{A}^{-1}$, as shown specifically in Figure 13.

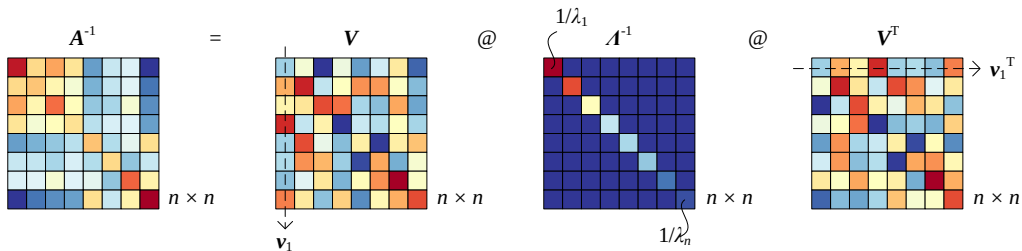


Figure 13. Spectral Decomposition of the $n \times n$ Real Symmetric Matrix $\mathbf{A}^{-1}$



Earlier in this book, the symmetric matrix you most commonly encountered was the Gram matrix. Section 4 of this chapter will explain the Gram matrix in detail.

Extracting Elements

At the end of this section, let's take a closer look at an interesting set of matrix multiplications in the equation above.

For example, the following matrix multiplication is equivalent to extracting the element in row 1, column 1 of the identity matrix I

$$\mathbf{e}_1^T \mathbf{I} \mathbf{e}_1 = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 1$$

As another example, the following matrix multiplication is equivalent to extracting the element in row 2, column 1 of the identity matrix I

$$\mathbf{e}_2^T \mathbf{I} \mathbf{e}_1 = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0$$



Please think about how to extract the remaining elements of the identity matrix I .

If the real symmetric matrix A can be spectrally decomposed to give $A = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$, the following matrix multiplication is equivalent to extracting the element in row 1, column 1 of the diagonal matrix $\mathbf{\Lambda}$ (not A)

$$\mathbf{v}_1^T \mathbf{A} \mathbf{v}_1 = \underbrace{\mathbf{v}_1^T}_{\mathbf{e}_1^T} \mathbf{V} \underbrace{\mathbf{A} \mathbf{V}^T}_{\mathbf{e}_1^T} \mathbf{v}_1 = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \lambda_1$$

This is equivalent to, in the orthonormal basis formed by the column vectors of $\mathbf{V}$, "setting upright" the real symmetric matrix A to obtain the diagonal matrix $\mathbf{\Lambda}$.

As another example, the following matrix multiplication is equivalent to extracting the element in row 2, column 1 of the diagonal matrix $\mathbf{\Lambda}$

$$\mathbf{v}_2^T \mathbf{A} \mathbf{v}_1 = \underbrace{\mathbf{v}_2^T}_{\mathbf{e}_2^T} \mathbf{V} \underbrace{\mathbf{A} \mathbf{V}^T}_{\mathbf{e}_1^T} \mathbf{v}_1 = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0$$



Please think about how to extract the remaining elements of the diagonal matrix $\mathbf{\Lambda}$.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

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Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

Q1. Please perform spectral decomposition on the following real symmetric matrices, and explain the decomposition results from a geometric perspective.

▶ $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Q2. Please modify LA_11_01_01.ipynb to visualize the spectral decomposition results from Q1.